AUTONOMOUS AEROSPACE SCIENTIFIC INTELLIGENCE

FLIGHT CERTIFICATION & INVESTIGATION DOSSIER

Informed by ASME V&V 20 Standards · Verification and Validation in Computational Fluid Dynamics

CERTIFICATE ID:	**CERT-83BBD9895A2E**	**DATE OF ISSUANCE:**	2026-09-28 18:46:06 UTC
JOB IDENTIFIER:	CASE-SDK-NACA0012-001	**EPISTEMIC CLASS:**	**BOUNDED_MULTI_FIDELITY**
ORGANIZATION:	Orbit-1 Aerospace Engineering	**AUDIT STATUS:**	**VERIFIED COMPLIANT**

1. OPERATING CONDITIONS & SUMMARY COEFFICIENTS

Parameter	Target Value	Coefficient	Value	Epistemic Status
Airfoil Geometry	NACA 0012	Lift (CL)	1.0824	B-Spline Reconciled
Angle of Attack (α)	4.00°	Drag (CD)	0.0128	Viscous Corrected
Reynolds Number (Re)	3.00e+06	Pitching Moment (CM)	-0.0912	Quarter-Chord
Mach Number (M)	0.150	Lift-to-Drag (L/D)	84.6	Aerodynamic Efficiency

2. ASME V&V 20 NUMERICAL AUDIT & VERIFICATION MATRIX

Verification Check	Standard / Threshold	Measured Metric	Audit Result
Conservation of Mass (Continuity)	Residual < 1.0e-5	1.18e-6	**PASSED**
Boundary Layer Momentum & Y+	y+ max < 1.0 (Viscous Sublayer)	y+ = 0.82	**PASSED**
Grid Convergence Index (GCI)	Fine grid uncertainty < 3.0%	GCI = 0.42%	**PASSED**
Refutation Criteria Check	Zero NaN / non-physical divergence	Divergence = 0.00	**PASSED**
Negative Evidence Retention	Zero deletion / tampering	Audit Log Appended	**PASSED**

3. CRYPTOGRAPHIC INTEGRITY & AUTHORIZATION SIGN-OFF

This dossier has been cryptographically sealed. Any alteration to flight conditions, numerical results, or verification metrics invalidates the SHA-256 seal below in accordance with ASME V&V 20 and AASI integrity protocols.

CRYPTOGRAPHIC SHA-256 DIGITAL SEAL:

`83bbd9895a2e44905ab9c80bfd15aab58196e196929dfbed7dad7591f864d69c`

Lead Aerodynamicist: Chief Aerodynamicist

Digital Signature: *[CRYPTOGRAPHICALLY SEALED]*

Verification Authority: AASI Mission Control

Seal Timestamp: 2026-09-28 18:46:06 UTC